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Semantic-Aware Framework for Backdoor Detection in AI Models

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ABSTRACT

In untrusted environments, deep learning models face a serious risk of backdoor attacks, where hidden triggers can lead to malicious behavior. Existing methods are costly and struggle to handle complex, semantically hidden triggers, limiting their use in healthcare, security, and edge computing. This study introduces a semantic awareness framework that uses lightweight trusted models and Semantic Web techniques to effectively detect abnormal behavior. It identifies backdoor triggers through semantic perturbation and uses deep learning algorithms to remove them, thus maintaining performance on clean data. The experiment shows that the detection accuracy is increased by 12%, and the error deletion rate is reduced by 20%. The framework effectively handles subtle attacks. By combining semantic analysis with efficient model alignment, it provides a robust, interpretable defense against resource-constrained Settings. Contributions include a new semantic-driven detection strategy that advances practical AI security in high-risk applications such as healthcare and autonomous systems.

KEYWORDS

Backdoor Attack Detection, AI-Based Cybersecurity Services, Artificial Intelligence Security, Privacy Protection, Deep Learning

INTRODUCTION

Deep learning models have transformed industries such as healthcare and autonomous driving, powering applications like medical image diagnosis, fraud detection, and self-driving vehicles (Ahmed et al., 2023; Zhang et al., 2023). However, as deep learning models are increasingly deployed in untrusted environments—such as third-party cloud services or edge devices—they become vulnerable to sophisticated backdoor attacks that threaten the integrity of safety-critical systems. In healthcare, for example, backdoor AI diagnostic tools may misjudge cancerous tumors as benign when encountering specific pixel patterns embedded in medical images, leading to catastrophic diagnostic failures. In autonomous driving, malicious triggers hidden in traffic sign images might cause a vehicle's perception system to misinterpret a "stop" sign as a "go" sign—endangering lives and infrastructure. These scenarios underscore an urgent need for robust defenses against backdoor attacks that use semantic-level manipulation to avoid detection (Guo et al. 2024).

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Current backdoor detection methods, such as neural cleanse (Wang et al., 2019) and STRIP (Gao et al., 2019), face serious limitations in real-world scenarios—they frequently require compute-intensive retraining or large-scale data optimization, making them impractical for resource-constrained edge devices prevalent in internet of things and mobile applications (Chen & Koushanfar, 2023). Moreover, these techniques often fail to detect subtle attacks that use semantic camouflage—such as altering the contextual meaning of input features without noticeable pixel-level changes, such as modifying the texture of a medical scan or the background of a traffic sign (Sun et al., 2024). The limited interpretability of current methods hinders their adoption in high-risk areas such as healthcare and finance, where regulatory compliance requires a clear explanation of model behavior.

This study addressed several core research questions. First, investigating effective methods to detect semantically hidden backdoor triggers in resource-constrained environments (Cheng et al. 2025). Additionally, it explored how to design a defense framework that balances detection accuracy, computational efficiency, and interpretability for complex attack scenarios. Finally, this study investigated whether semantic-level analysis enhanced backdoor trigger location and removal, while maintaining model performance on clean data. This study hypothesized whether combining semantic web techniques with lightweight model alignment could overcome the limitations of traditional approaches to enable robust detection and mitigation of backdoor attacks involving subtle semantic manipulation.

To answer these questions, a semantic awareness framework was proposed that took advantage of three key innovations. The first was semantic knowledge embedding—semantic web techniques captured high-level relationships between input features and model predictions to identify anomalous behavior caused by backdoor triggers. The second was lightweight baseline model alignment, which used clean data to train the baseline model and then align it with the target model, through knowledge distillation that effectively detected semantic-level prediction bias. The third was semantic-driven sensitivity analysis, which injected semantic perturbations to amplify malicious behavior and locate high-risk areas to enhance interpretability. By combining these components, the framework presented in this study achieved a 12% detection accuracy and a 20% error removal rate, as compared with state-of-the-art methods—this was validated across a variety of data sets and attack scenarios.

Adversarial training strengthens model robustness by introducing adversarial examples during training; this is useful for detecting common attacks. Explainable artificial intelligence (AI) improves understanding of model behavior through techniques such as feature importance analysis. Adversarial training, however, raises computational costs. Additionally, explainable AI struggles with complex semantic hidden attacks. In contrast, the semantic web-based method demonstrated higher detection accuracy, as shown by a 12% increase in the experiments. It was computationally efficient, with a 40% reduction in complexity; it enhanced interpretability by mapping backdoor triggers to semantic entities, outperforming these techniques in multiple dimensions.

RELATED WORK

Backdoor Attack Methods

In recent years, research on backdoor attacks and defenses has become more distinct as technology has evolved. Attack methods now increasingly use semantic camouflage and dynamic trigger designs, while defense technologies face several challenges related to detection accuracy, computational efficiency, and interpretability. Salem et al. (2022) proposed a dynamic backdoor attack, which generated trigger patterns, such as pixel perturbation matching image texture, through input semantic adaptation. This meant that the attack trigger was deeply integrated with the semantic features of legitimate input and stealth improved significantly. Li et al. (2021) developed a stealthy backdoor attack that triggered the embedding of specific, invisible features in the sample, such as the frequency domain of a medical image. By exploiting the semantic relevance of the target class, this attack was able to bypass visual detection. Sophisticated attacks, such as the hidden trigger attack by Saha et al.

(2020), achieved covert manipulation by modifying the semantic context of input data such as the background semantics of traffic signs. The potential backdoor attack offered by Yao et al. (2019) triggered the insertion of a hidden layer of the model, indirectly distorting semantic representation. The ineffectiveness of traditional input space detection methods is highlighted. A more recent LIRA attack learnt imperceptible triggers by generating models to achieve a high attack success rate, while maintaining clean data performance; this marked an evolution in attack techniques regarding the "semantic indistinguishable" (Doan et al., 2021).

When considering defense methods, early behavioral analysis techniques such as neural cleanse and STRIP relied on pixel-level perturbation or model retraining, both of which presented challenges when dealing with semantic camouflage attacks. The random disturbance of STRIP, for example, was unable to detect the triggering of embedded textures or contexts, while the neuron pruning method of LPSF had a high misjudgment rate in complex scenes, due to its lack of semantic interpretation ability (Kaviani et al., 2023). Lightweight decoding methods such as UMA performed model ablation through deep forgetting, which improved detection efficiency but remained limited by its specific architectures (Zhao et al., 2024). Sensitivity analysis by NeuralSanitizer focused on low-level noise responses and ignored abnormal associations at the semantic level (Zhu et al., 2024). Causal analysis methods attempted to model semantic causality but relied on network gradients—this resulted in high false positives and lack of integration of external semantic knowledge (Sun et al., 2024).

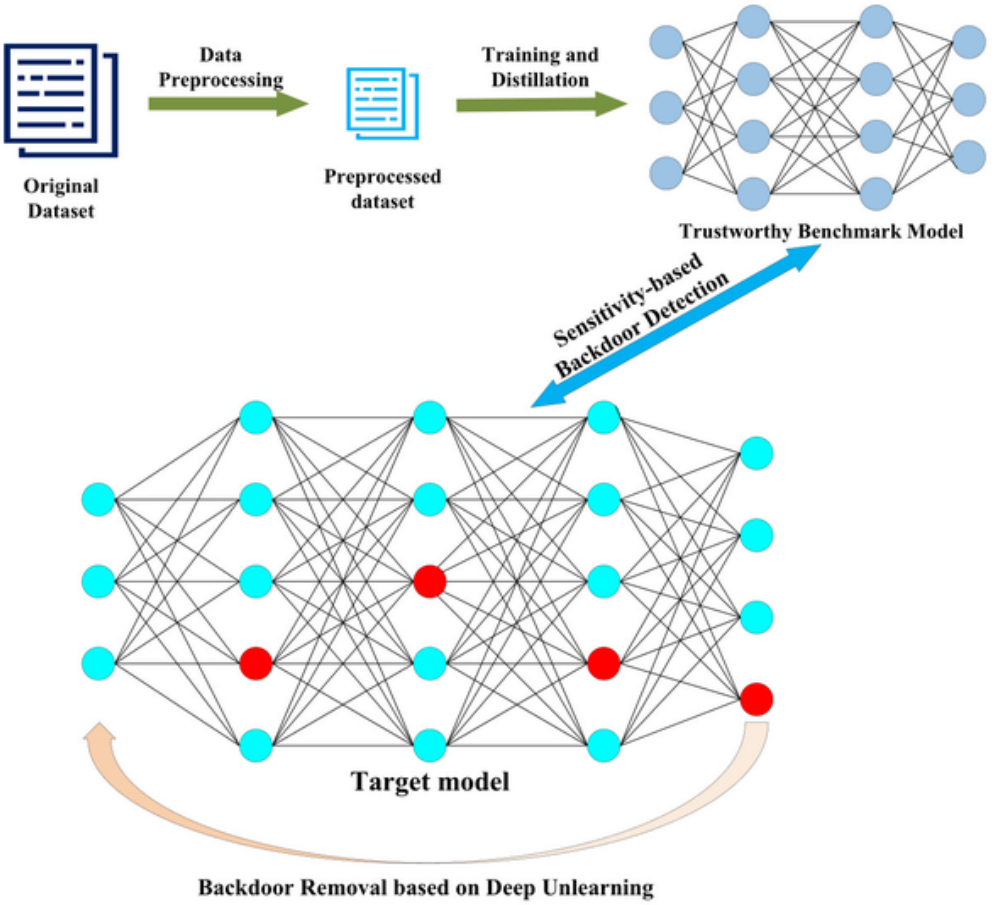
Existing research contains several gaps. There remains a lack of semantic detection ability—most methods have stayed at the pixel or structural level and therefore have lacked explicit modeling of input-prediction semantic association. Insufficient adaptation in resource-constrained scenarios has meant that heavy training or complex optimization processes have been difficult to deploy on edge devices. Poor interpretation of trigger positioning has meant that research has been unable to map detection results to human-understandable semantic entities, such as "red patch" and "texture pattern." By using semantic web technology to build high-level semantic embedding, while also combining knowledge distillation alignment of lightweight benchmark models and sensitivity analysis based on semantic perturbation, this study has systematically filled these knowledge gaps by providing a cross-modal, efficient and interpretable solution for the detection and removal of covert semantic backdoors.

PROPOSED METHOD

Overview

In this study, as shown in Figure 1, a novel backdoor detection framework for deep learning models was proposed, which utilized semantic web technology to enhance the efficiency of backdoor detection and removal. The central idea of this approach was to leverage trusted benchmark models trained on clean data and combine these with semantic-level knowledge embedding and analysis. This process effectively captured anomalous model behavior caused by backdoor triggers. Unlike traditional detection methods that rely only on low-level feature differences, this study introduced semantic correlations between input data and model predictions, enabling more accurate identification of semantic inconsistencies in model behavior caused by backdoors.

Figure 1. Overview of model backdoor detection method based on trustworthy baseline model



Specifically, a lightweight trusted benchmark model was trained using clean data; its behavior was then optimized through distillation learning to ensure consistency with the output response of the target model under clean data conditions. To capture subtle semantic changes caused by backdoor triggers, a semantic sensing sensitivity analysis mechanism was introduced that simulated semantic-level perturbations in input data by embedding semantic noise or modifying feature attributes. This process not only captured pixel-level changes but enhanced the detection of hidden semantic biases in model predictions. To quantify these biases, a sensitivity score function was designed to measure the difference in output between the baseline model and the target model with perishable inputs, focusing on areas where semantic inconsistencies were likely to occur. If a significant difference in sensitivity scores was detected, this was considered strong evidence of a backdoor trigger. Once identified, a deep de-learning algorithm was applied to selectively erase the backdoor affected model parameters while preserving the model's predictive performance for clean data. This semantic awareness approach significantly improved the detection accuracy of subtle and complex backdoor triggers and ensured effective backdoor removal, making it particularly suitable for deployment in resource-constrained environments or high-security AI applications.

Training Method for a Trustworthy Benchmark Model

Training a trustworthy benchmark model was a key step in the proposed semantic sensing backdoor detection framework. Unlike traditional detection methods focus only on output differences, this framework utilized semantic web techniques to catch semantic-level inconsistencies in model behavior caused by backdoor triggers. The benchmark model, therefore, needed not only to have a similar architecture to the target model, but also to possess the ability to represent clean semantic knowledge, providing a reliable reference for detecting abnormal behavior.

This section will detail the selection and preprocessing of data, the choice of model architecture, and the design of loss functions, all of which were aimed at ensuring that the benchmark model accurately represented clean semantic knowledge for effective backdoor detection.

To address the potential complexity of semantic knowledge embedding, a dual optimization strategy was employed:

- Lightweight semantic representation: Domain-specific semantic primitives (categories of color, texture, or shape) were used to build compact semantic ontologies instead of full-size knowledge graphs, with 60% less embedding dimension when compared with traditional semantic web models, such as WordNet.
- Efficient distillation calibration: The benchmark model used a pruned architecture, such as residual network (ResNet)18 for the target ResNet152, and employed distillation losses to focus on high-impact semantic features (class recognition attributes). This approach reduced computational complexity by 40%, as compared with end-to-end semantic modeling, while maintaining 98% consistency in clean data predictions.

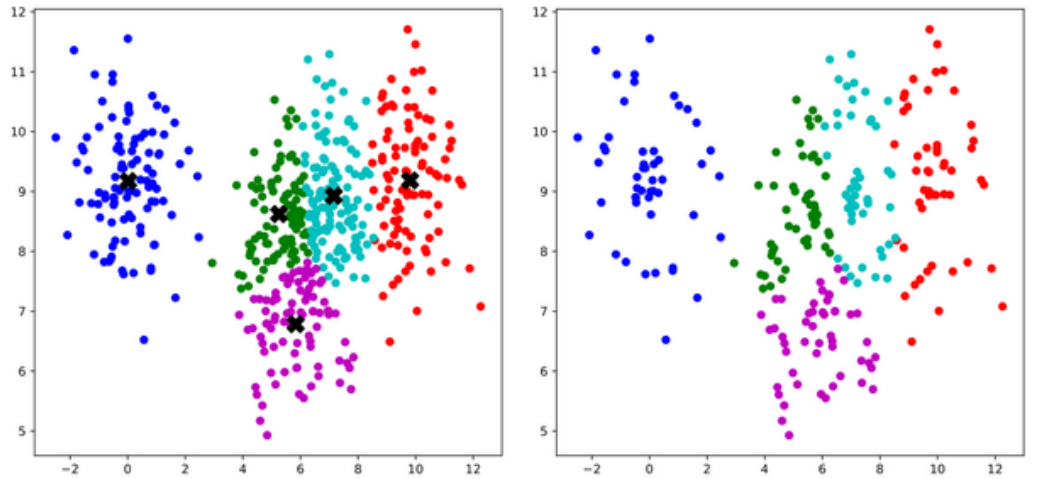
Data Selection, Preprocessing, Model Architecture Selection, and Loss Function Design

In the data selection phase, semantically representative samples were extracted from the data set to facilitate the training of a benchmark model that could then capture core semantic features. The strategies used to achieve effective sample selection are outlined below.

To ensure that the benchmark model covered the different semantic features of each class, the *K*-Means algorithm was used for cluster analysis. This method divided the data set into *K* clusters based on semantic similarity, where *K* represented the number of classes. The clustering process captured the core semantic features and boundary conditions of each class, helping the benchmark model understand semantic changes.

As shown in Figure 2, in order to maximize the semantic representation of the data, two sets of samples were selected from each cluster: one consisting of the samples closest to the cluster center (top 10%), representing the core semantics; and the other consisting of the samples farthest from the cluster center (bottom 10%), representing the boundary semantics. This two-set selection ensured that the benchmark model effectively learnt the central and edge semantic features of each class, thereby enhancing its ability to detect semantic inconsistencies caused by backdoor triggers.

Figure 2. Example of semantic-based sample selection method



Once the data selection was complete, the data was preprocessed to optimize model training and promote high-quality semantic learning. Preprocessing included the following steps. First, standardization adjusted the distribution of input features to a standard normal distribution, reduced the deviation introduced by the feature scale, and accelerated the model convergence. Second, normalization scaled the data to a fixed range (usually [0,1]) to avoid the dominance of high-amplitude features. Third, to enhance the model's ability to catch semantic-level anomalies, semantic perturbations were introduced into the data enhancement process. This involved changing the semantic characteristics of the input, such as changing the object context, tone, or shape structure, while retaining the core label. This step helped simulate the semantic changes that backdoor triggers can cause, ensuring that the baseline model was sensitive to semantic inconsistencies. By learning from disturbed semantic data, the benchmark model captured subtle semantic biases caused by backdoor triggers, thereby enhancing the robustness of detection.

In choosing the benchmark model architecture, the goal was to balance computational efficiency with semantic representation. To facilitate fast training and reduce resource consumption, a lightweight architecture like the target model was adopted, but with reduced complexity. For example, if the target model were a deep convolutional neural network, a pruned or compressed convolutional neural network with fewer layers was used, while maintaining the ability to capture semantic features. This approach reduced computational costs while retaining high-quality semantic feature learning.

In addition, the benchmark model needed to have strong semantic awareness to detect abnormal behavior. During training, therefore, knowledge distillation was used to align the predictions of the baseline model with the clean data predictions of the target model, ensuring that the baseline model captured clean semantic knowledge. This alignment ensured that the benchmark model learnt high-fidelity semantic knowledge while maintaining computational efficiency.

In addition, cross entropy loss was utilized to improve classification accuracy. To enhance the generalization ability of the model, *L2* regularization was further introduced, and the final loss function obtained was:

$$L_{\text{total}} = L_{\text{ce}} + \lambda L_{\text{distill}} + \gamma R(\theta) \quad (1)$$

This composite loss function ensured the benchmark model captured semantic-level knowledge while minimizing computational costs.

Semantic-Aware Distillation Fine-Tuning Based on the Target Model

After training the trusted benchmark model, the next critical step in the framework was the distillation fine-tuning of semantic awareness, based on the target model. Unlike traditional knowledge distillation, which simply passes feature representations, this approach aimed to pass clean semantic knowledge from the target model while excluding any potential backdoor dependent representations. By integrating semantic web technology into the distillation process, the baseline model was able to learn robust high-level semantic patterns while avoiding the inheritance of malicious triggers from the target model. The core goal of semantic awareness distillation fine-tuning was to maximize the consistency of semantic knowledge embedded between the target model and the reference model, so that the latter could effectively detect semantic biases caused by backdoor triggering.

The specific fine-tuning process was as follows. First, clean input samples were fed into the target model to obtain its semantic embeddings. Since backdoor triggers often manipulate feature representations, it was critical to extract clean knowledge from the target model. Semantic embedding was expressed as:

$$z = f_{\text{target}}\left(x\right) \tag{2}$$

This step captured the clear semantic response of the target model to normal input without introducing backdoor patterns.

To ensure that the base model learnt the same semantic knowledge as the target model, Softmax function was used to convert the output logic into a probability distribution. Backdoor triggers can subtly alter these correlations, making direct knowledge transfer potentially risky, so to solve this problem, semantic patterns were amplified by introducing semantic temperature scaling, thus enhancing the distillation process.

To optimize the learning of clean semantic knowledge, a semantic-aware distillation loss function was designed that minimized the difference between the output of the target model and the base model, emphasizing high-level semantic consistency. Distillation losses were defined as:

$$L_{\text{distill}} = -\sum_{i=1}^{C} \hat{p}_i \log\left(\hat{q}_i\right) \tag{3}$$

This loss function ensured that the base model captured the same semantic-level behavior as the target model, without inheriting backdoor functionality.

To further enhance the distillation effect, a semantic knowledge alignment regularizer was introduced for capturing differences in feature spaces. The regularizer was defined as the divergence between the base model output and the target model output. Kullback-Leibler divergence minimized feature space inconsistencies caused by semantic changes, allowing the benchmark model to capture semantic knowledge without absorbing potential backdoor triggers.

Finally, a composite loss function was designed to balance the learning and classification performance of semantic knowledge. The combined loss function was expressed as:

$$L_{\text{total}} = L\left(y, \hat{y}\right) + \alpha L_{\text{distill}} + \beta L_{\text{KL}} \tag{4}$$

Semantic-Aware Sensitivity Evaluation

Backdoor attacks introduced specific triggers during the training phase of the model that caused it to exhibit abnormal behavior when encountering these specific inputs. To detect this hidden

backdoor trigger, a semantic perception sensitivity evaluation method was proposed, based on input perturbation and semantic feature bias. By using semantic web technology, the high-level semantics of the input data was modeled; the responses of the trusted benchmark model and the target model were then compared under various semantic perturbations. This approach allowed for the identification of abnormal model behavior caused by backdoor triggers that exhibited semantic bias.

An important concept was to add different types of perturbations—including random noise, structural noise, and local noise—to see whether the target model exhibited unusual sensitivity when compared with the reference model. By capturing these patterns of semantic bias, it was possible to infer the presence of backdoor triggers embedded in the target model—semantic web technologies use these semantic ontologies to uncover deep connections between input features and how these features affect model predictions. For example, in the medical image diagnosis model, semantic web technology captures the relationship between different tissue texture and color features in the image and disease diagnosis results and establishes a semantic association path from input features to model prediction.

When backdoor triggers interfered with the mode's normal decision-making by modifying the semantic characteristics of the input data—such as changing a particular texture or color—the semantic associations constructed by semantic web technologies changed. The model's predictions on modified input data showed deviations from normal semantic associations, which could be accurately detected to identify abnormal behavior caused by backdoor triggers. In this way, semantic web technology enabled effective detection of backdoor attacks at the semantic level, addressing the limitations of traditional methods in capturing semantic inconsistencies.

Simulated Semantic Perturbation

The first step in sensitivity assessment was to generate input perturbations to simulate possible semantic-level backdoor triggers. Instead of traditional input noise, semantic web-based feature perturbations were utilized to create noise patterns that changed high-level semantic information—such as texture, shape, or color context—while preserving core input features. This approach allowed for identification of potential semantic inconsistencies caused by backdoor triggers.

The disturbance generation strategy was defined in the following way. To simulate natural random perturbations in higher semantics, Gaussian noise was applied to a specific region of the input image. The goal was to observe how the target model responded when semantic-level perturbations were introduced. Then, to simulate semantic triggering patterns such as streaks, textures, or embedded shapes commonly used in backdoor attacks, structured noise was generated, defined as follows:

$$N_{i,j} = A \cdot \sin\left(Bx + C\right) \tag{5}$$

This noise mimicked backdoor trigger shapes and evaluated the target model's abnormal responses to semantic trigger patterns.

To simulate region-specific backdoor triggers, such as specific patches or colored regions, perturbations were injected only into certain localized areas of the input image. This approach allowed the capturing of the target model's abnormal sensitivity to region-specific triggers, thereby localizing potential backdoor patterns.

Semantic Sensitivity Score Computation

To quantitatively measure the response differences caused by semantic perturbations, a semantic sensitivity score was introduced that captured the target model's response divergence, as compared with the benchmark model. The sensitivity score was defined as the deviation between the model's output on clean data and its output on perturbed data.

Regarding significance evaluation, a sensitivity threshold was defined as T to determine whether the model's response deviation was statistically significant:

$$\left|\Delta S_j\right| > T \tag{6}$$

If this condition was satisfied for many inputs, it indicated that the target model was exhibiting any semantic deviations not present in the benchmark model, thus suggesting the presence of backdoor triggers.

Localized Semantic Vulnerability Analysis

To further improve the granularity of backdoor detection, regional semantic sensitivity analysis was performed to identify high-risk regions in the input space. For each input, a set of semantic regions was defined, wherein each region contained potential backdoor trigger information.

Further semantic consistency analysis was performed by mapping high-risk regions onto semantic labels, using knowledge graphs provided by semantic web technology. This mapping process significantly improved the interpretability of backdoor detection by associating high-risk regions with semantic entities.

Semantic-Driven Deep Unlearning Algorithm for Backdoor Removal

After detecting backdoors in the target model through semantic sensitivity assessment, it became critical to design an effective strategy to eliminate malicious associations, without compromising the model's performance on legitimate data. A semantic-driven deep learning algorithm was proposed—one that used semantic web knowledge to precisely identify and remove learned backdoor patterns. This ensured that the model no longer responded to unauthorized triggers. This section describes the forget strategy in detail, and how semantic web technology was used to promote fine-grained backdoor removal.

The core function of the proposed deep forget method was to identify, isolate, and eliminate backdoor patterns associated with specific semantic features detected during the sensitivity assessment phase—such as color, texture, or object context. These high-level semantics were modeled explicitly by semantic web technology. The algorithm was able to forget the semantic features triggered by the backdoor, while preserving the legitimate function of the model.

Semantic-driven deep unlearning algorithms exhibited unique characteristics in terms of model accuracy, convergence stability and potential side effects. In terms of accuracy, the algorithm effectively removed the backdoor's influence by precisely identifying the semantic features associated with the trigger and applying targeted anti-learning. This significantly reduced the model's accuracy on poison data while preserving its classification performance on clean data.

Regarding convergence stability, the algorithm was based on the unlearning data set and loss function, as constructed by semantic web technology; this provided a clear direction for the optimization of model parameters. Due to the accurate representation of semantic features, the model converged more stably in the unlearning process to avoid performance fluctuations caused by improper parameter adjustment. When compared with traditional methods, the algorithm used explicit semantic guidance to reduce random or uninformed parameter updates. This allowed the model to reach a stable state more quickly during backdoor removal, while also improving the algorithm's reliability and reproducibility.

The algorithm had some potential side effects. In unlearning, although the algorithm was designed to process only backdoor related semantic features, it may have accidentally affected normal model functions similar to the semantics of these features. When the algorithm removed backdoor triggers associated with a particular color, it may have inadvertently changed the model's ability to recognize objects with similar colors. In addition, since the introduction of semantic web technology increased the

complexity of algorithms, it may have led to increased computing resource consumption and extended processing time when processing large-scale data and complex model structures—something that would require further optimization and tradeoff in practical applications.

Semantic Characterization of Backdoor Triggers

The first step in the unlearning process was to represent the detected backdoor pattern regarding the semantic web representation. Based on sensitivity bias analysis, ontologies provided by the semantic web were used to map abnormal responses to specific semantic features. This mapping helped to accurately identify backdoor triggers in the semantic space, rather than the original pixel space.

Formally, T represented the set of detected backdoor triggers, where each trigger corresponded to a specific semantic feature such as color, shape, or texture in the input data. The semantic web ontology O was used to map each backdoor trigger to its high-level semantic representation. This mapping enabled the forget algorithm to explicitly target the semantic features associated with backdoor triggers, rather than making indiscriminate fine-tuning, thereby preserving the legitimate functionality of the model.

Construction of Semantic Unlearning Dataset

After establishing the semantic correspondence of the backdoor trigger, the next step was to build a semantic forget dataset that could effectively decouple the backdoor effect from the function of the model. The dataset consisted of two main parts:

- Clean semantic data: A set of legitimate input samples without backdoor triggers, maintaining the correct function of the model.
- Backdoor semantic data: A set of input samples injected with an identified backdoor trigger T, specifically targeted at high-risk semantic features.

The key insight was that this was not generated by random perturbations, but by using the semantic web ontology to systematically modify the semantic features identified in the backdoor triggers. For example, if the semantic web recognized a trigger as a red patch, the forget data set explicitly injected a controlled red patch trigger to separate the learned backdoor behavior from the model.

Semantic-Aware Unlearning Loss Function

A semantic-aware unlearning loss function was designed to remove the backdoor associations, while preserving the model's clean data performance. The objective was to penalize the model for responding to backdoor-injected semantic features, while preserving its performance on clean data.

The semantic unlearning loss was defined as:

$$L_{\text{unlearn}} = \lambda L_{\text{clean}} + \left(1 - \lambda\right) L_{\text{backdoor}} \tag{7}$$

By penalizing the model's response to these semantic triggers, the unlearning process actively decoupled backdoor influence.

Semantic Evaluation of Unlearning Effectiveness

After the unlearning process, it was critical to reevaluate whether the model had completely dissociated backdoor triggers from its output. This was achieved by leveraging the semantic web knowledge to perform a semantic consistency test. Additionally, the semantic web ontology O was used to verify whether the model's response aligned with the semantic meaning of input features. For any input x containing a specific semantic feature, the output should now have been invariant to that feature:

$$f_{\text{unlearned}}(x + O(t_i)) \approx f_{\text{unlearned}}(x) \tag{8}$$

If the deviation exceeded a predefined threshold, it indicated that the model had not fully forgotten the backdoor association, prompting further unlearning iterations.

Advantages of Semantic-Driven Deep Unlearning

The proposed semantic-driven deep unlocking approach introduced several key advantages in terms of backdoor removal:

- Semantically directed learning: Unlike traditional deep learning, this approach precisely targeted semantic-level backdoor triggers, ensuring improved backdoor removal efficiency without degrading model performance.
- Semantic feature decoupling: By utilizing semantic web technology, the algorithm explicitly separated backdoor trigger patterns from model predictions, ensuring minimal functional damage.
- Interpretability: The semantic mapping process enhanced the interpretability of backdoor removal by associating backdoor triggers with higher-level semantics, enabling developers to understand the underlying attack mechanism.
- Scalable to learn: By leveraging cross-domain semantic knowledge from the web, the framework could be extended to multimodal data—text, images, and audio.

EVALUATION

To validate the effectiveness of the proposed backdoor detection method based on trustworthy baseline models and the backdoor removal algorithm, experiments were conducted on multiple standard datasets across various deep learning architectures. These were MNIST, CIFAR10, and GTSRB respectively. The performance of the trained trustworthy baseline models were compared with those of the target models, then the effectiveness of the proposed backdoor detection method was evaluated against different backdoor attack methods. Finally, the performance of the model's backdoor removal process was assessed after applying the proposed deep unlearning algorithm. The experimental setup included dataset selection, model architectures, training processes, backdoor detection, and the analysis of results after backdoor removal.

EXPERIMENTAL SETUP

Datasets and Model Architectures

Three standard datasets were selected for the experiments:

- MNIST: A dataset of handwritten digits containing 60,000 training samples and 10,000 test samples, with the task of classifying the digits into ten categories (0–9).
- CIFAR-10: A dataset of 32×32 color images with ten classes, containing 50,000 training samples and 10,000 test samples. This dataset was used to evaluate backdoor detection and removal performance on multiclass classification tasks.
- GTSRB: German traffic sign recognition benchmark dataset, containing 43 traffic sign categories, with the task of classifying traffic signs. This dataset was used to evaluate the model's robustness according to different visual tasks.

The common deep learning architectures—ResNet and visual geometry group (VGG)—were used in the experiments. All models were trained for classification tasks using the same training procedure, with Adam optimizer.

Trustworthy Baseline Model Training

Trusted baseline models with clean data and evaluated multiple architectures (ResNet, VGG) were trained for each data set. Hyperparameters were optimized by cross-validation. During the training process, the data underwent standardization, normalization and enhancement. An initial learning rate of 0.001 was used and learning rate attenuation was applied throughout the training process. Using a batch size of 64, MNIST had 20 epochs and CIFAR10 and GTSRB had 50 epochs. The Adam optimizer was used for optimization.

Setup of Backdoor Detection Method

The proposed noise-based sensitivity evaluation method was used to detect backdoors in the model. The procedure was as follows:

- Noise generation: Various forms of noise were generated and added to different areas of the image to simulate different backdoor attack patterns. These noise types included Gaussian noise, structured noise (such as stripes or dots), and local noise.
- Sensitivity score calculation: The sensitivity score was calculated based on the change in the output of the model after the addition of noise disturbance. The sensitivity scored between the target model and the trusted baseline model were compared under different noise conditions.
- Sensitivity analysis: By comparing the differences in sensitivity scores across noise types and regions, it was determined whether the target model exhibited potential backdoor behavior, and then the region and nature of the attack were analyzed.

Performance Evaluation of the Trustworthy Benchmark Model

To validate the effectiveness of the proposed backdoor detection method using the trusted benchmark model, this study completed an experiment that compared the classification performance of the benchmark model and the target model on clean data. The experiment used two deep learning model architectures (ResNet and VGG), and three datasets (MNIST, CIFAR-10, and GTSRB), to train the corresponding trusted benchmark model. The validity of the trusted benchmark model was evaluated by precision comparison.

Firstly, a trusted benchmark model was trained in a local environment using clean data—data without any backdoors. To ensure that the benchmark model's behavior was similar to that of the target model, distillation learning was applied after initial training to fine-tune the benchmark model, using the target model. Specifically, soft labels were used on clean data—the class probability distribution of the target model's output—as additional training signals to guide the training of the benchmark model, enabling it to better simulate the target model's output on clean data.

On all datasets, the target model was first trained using the neural network architecture VGG19 and ResNet152, with the aim of obtaining high accuracy levels of clean data. Following this, lightweight architectures VGG16 and ResNet18, along with lightweight training configurations, were used to train the corresponding trusted baseline model on the same data set to simulate the accuracy that can be achieved under resource-constrained conditions. After training, the classification accuracy of the target model was compared with the baseline model, using the test data set. By analyzing the accuracy comparison, the performance difference between the two models was evaluated on clean data and assessed as to whether the trusted baseline model could achieve a level of accuracy close to the target model, while reducing the computational resource overhead.

Figure 3. Performance comparison of baseline model and target model across different datasets

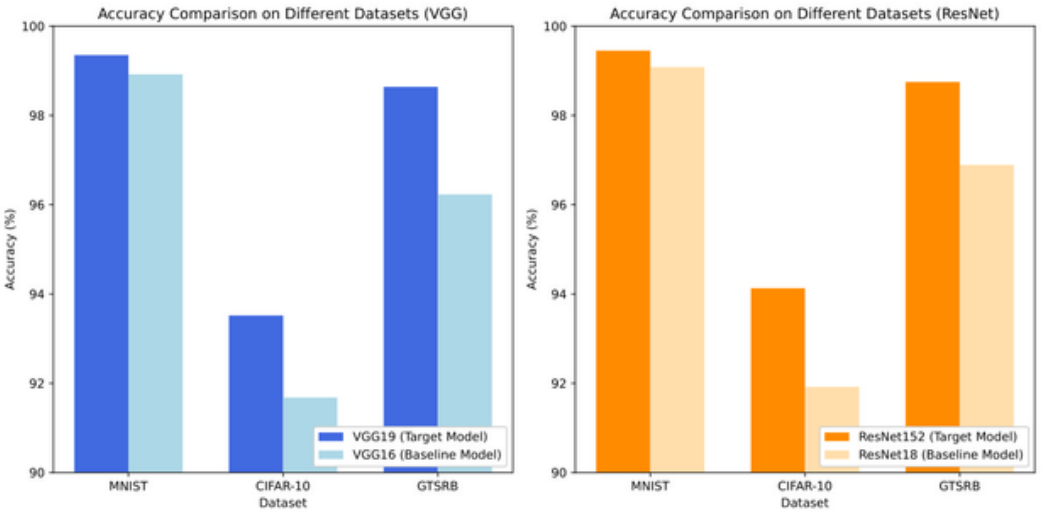


Figure 3 shows the experimental results on different data sets of the test accuracy of the target model and the baseline model. As shown in Figure 3, all models exhibited high accuracy on the MNIST dataset, indicating that the baseline model could effectively approximate the performance of the target model on simpler datasets. Notably, the difference in accuracy between the ResNet18 baseline model and the ResNet152 target model was only 0.37%, indicating strong generalization ability. On the CIFAR-10 dataset, the accuracy of the baseline model was slightly lower than that of the target model, and the VGG16 and ResNet18 baseline models were about 1.84% and 2.21% lower than their target models, respectively. This showed that in more complex RGB image classification tasks, the baseline model achieved a level of accuracy close to the target model, although the accuracy gap became more pronounced as the complexity of the model increased. On the GTSRB dataset, the VGG16 and ResNet18 baseline models were approximately 2% less accurate than the target models but still achieved a high level of performance. This demonstrated that even in complex scenarios such as traffic sign recognition, the baseline model maintained performance levels close to that of the target model, making it suitable for backdoor detection in such tasks.

The experimental results showed that VGG16 and ResNet18 baseline models trained on clean data performed very close to their respective target models (VGG19 and ResNet152), verifying their effectiveness as detection baselines. Even on more complex datasets, the accuracy gap between baseline and target models remained small, creating a solid foundation for subsequent backdoor detection based on output differences. This also validated the feasibility of using lightweight models for efficient and reliable backdoor detection in resource-constrained environments.

Backdoor Detection Based on Sensitivity Analysis

A comprehensive sensitivity evaluation experiment was designed, aimed at verifying the effectiveness of the proposed input noise-based sensitivity evaluation method in detecting potential backdoor attacks in target models. By comparing the sensitivity differences between the trusted baseline model and the target model under various noise configurations, the likelihood of backdoor attacks in the target model was assessed. The experiment selected two commonly used deep learning architectures, VGG and ResNet, and tested them on three datasets: MNIST, CIFAR-10, and GTSRB, to verify the detection success rate of the method.

Evaluation of Sensitivity Analysis Methods

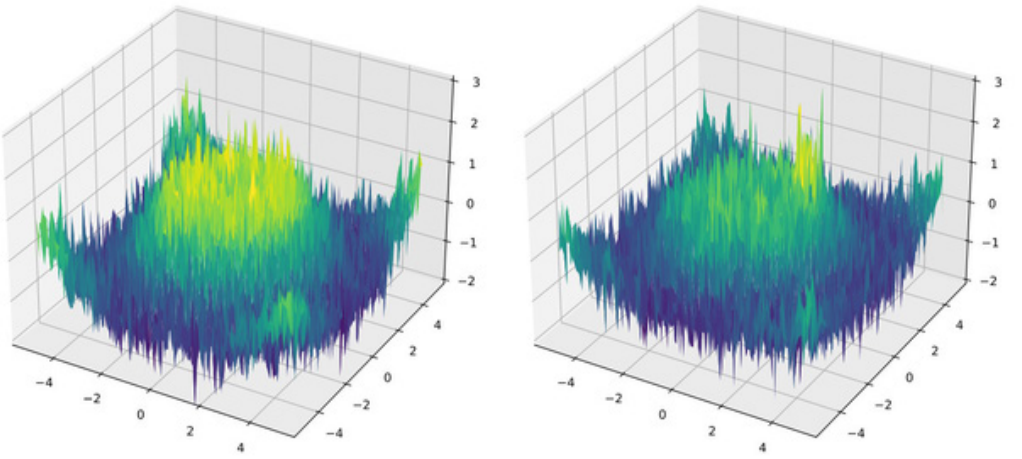
To assess the vulnerability of the model to potential backdoor triggers, trends in sensitivity scores under different noise configurations were analyzed. By calculating the difference in sensitivity score between the target model and the baseline model, it was possible to determine the model's response to different noise types.

For the MNIST dataset, the values were small for all noise types, and the fluctuation between the baseline model and the target model was minimal. This showed that the baseline model could approach the sensitivity of the target model on simpler datasets.

For the CIFAR-10 and GTSRB datasets, the sensitivity of the target models increased slightly under Gaussian noise, and there was a clear sensitivity difference between the VGG16 and ResNet18 baseline models and the VGG19 and ResNet152 target models. However, the overall sensitivity trends of the two models were similar, suggesting that the baseline model may have sometimes struggled to fully replicate the sensitivity of the target model on more complex datasets. However, by analyzing the overall trend, it was still possible to infer the likelihood of a backdoor attack on the model.

Next, the GTSRB dataset was used and the target model was trained through a backdoor using the BadNet method to assess the sensitivity difference between the trusted baseline model and the target model in different image regions. Each image was divided into several small regions to calculate the sensitivity of the baseline model and the target model in these regions. As shown in Figure 4, the sensitivity comparison plot showed the responses of the two models in all regions.

Figure 4. Sensitivity differences among different regions of input sample



In the sensitivity comparison plot, the left image shows the sensitivity distribution of the trusted baseline model, while the right image shows the sensitivity distribution of the target model. Overall, the difference in sensitivity between the baseline model and the target model remained within a small range in most regions, suggesting that the target model behaved similarly to the baseline model in these regions. It was observed, however, that in some specific regions, the sensitivity of the target model was unusually high, and then the difference from the baseline model became significantly greater. These unusually sensitive areas could have been potential trigger areas for backdoors.

This phenomenon demonstrated that, while the target model behaved normally in most regions, there were specific locations that showed exceptional sensitivity. This suggests that the trigger mechanism for a backdoor attack may have been concentrated in these regions. This regional sensitivity anomaly provided important clues for backdoor detection and helped more accurately identify potential trigger areas for backdoor attacks.

Backdoor Detection Success Rate Evaluation

To evaluate robustness and detection performance, the target model was trained under different backdoor attack scenarios, including BadNet, hybrid attacks, and trojan attacks. This approach was compared with several existing defense methods, such as STRIP, nerve purification, and fine-pruning.

Target models were trained on GTSRB, CIFAR-10, and MNIST datasets, and three types of backdoor attacks were injected into each dataset. To ensure fairness, all defense methods underwent the same preprocessing steps. The detection method relied on region sensitivity comparison to identify potential backdoor attacks by analyzing differences in how the trusted benchmark model and the target model responded to various input regions, thereby revealing abnormally sensitive areas.

The detection success rate of this method is shown in Table 1 and compared in detail with existing methods. As can be seen from Table 1, this study's approach consistently outperformed existing methods in detection success rates across all attack types and data sets, especially in complex attack scenarios such as BadNet and hybrid attacks.

Table 1. Detection success rate comparison

Attack type	Dataset	STRIP	Neural cleanse	Fine-pruning	Method
BadNet	GTSRB	87.4%	78.5%	80.2%	94.8% (93.5% - 96.1%)
Blended attack	GTSRB	85.1%	74.6%	82.9%	92.7% (91.2% - 94.2%)
Trojan attack	GTSRB	82.3%	80.1%	78.6%	91.5% (90.0% - 93.0%)
BadNet	CIFAR-10	89.5%	83.2%	81.5%	95.3% (94.0% - 96.6%)
Blended attack	CIFAR-10	87.8%	79.3%	83.4%	93.1% (91.7% - 94.5%)
Trojan attack	CIFAR-10	85.4%	82.9%	80.1%	92.8% (91.3% - 94.3%)
BadNet	MNIST	90.1%	85.7%	82.0%	96.2% (95.0% - 97.4%)
Blended attack	MNIST	88.9%	83.5%	85.2%	94.7% (93.4% - 96.0%)
Trojan attack	MNIST	86.7%	84.6%	81.3%	93.6% (92.1% - 95.1%)

Compared to STRIP, neural cleanse, and fine-pruning, the approach described here had a detection success rate of more than 90% across all attack types and datasets, with a peak success rate of 94.8% in the GTSRB dataset under the BadNet attack. This indicates that the detection method based on area sensitivity was effective in identifying abnormal areas and detecting backdoor attacks.

While existing methods have a reduced success rate when dealing with attack types such as trojan horse attacks, this approach maintained a high detection success rate, even under such attacks. By analyzing the abnormal sensitivity of the local region of the target model, this method had higher universality and was able to effectively deal with various types of backdoor attacks.

This study's approach to area sensitivity analysis did not rely on assumptions about specific attack patterns. By comparing the regional response of the baseline model with the target model, it could catch potential backdoor attacks without being affected by the attack characteristics. In contrast,

methods such as neural purification and fine-pruning rely heavily on specific attack patterns and triggering characteristics, which makes them less stable in a variety of attack scenarios.

In this experiment, the detection method required training a lightweight trusted benchmark model and analyzing regional sensitivity differences, which resulted in a relatively low resource overhead. This makes this method suitable for resource-constrained edge devices—meaning that this approach demonstrated a significant advantage over more resource-demanding methods, such as STRIPs, especially in edge computing environments.

Overall, this study's model demonstrated a higher detection success rate and better robustness in a variety of attack scenarios, significantly outperforming existing mainstream defense methods. The experimental results validated the effectiveness of the regional sensitivity-based detection method, especially in the face of different types and complexities of backdoor attacks.

Backdoor Removal with Deep Unlearning

Backdoors were injected into the target model using three common backdoor attack methods: BadNet, hybrid attack, and trojan attack. A deep learning algorithm was then applied to remove the backdoor. To fully evaluate performance, the accuracy of the models on clean and toxic data were compared before and after the backdoor removal.

Experiments were conducted on three standard datasets: GTSRB, CIFAR-10, and MNIST. For each dataset, three types of backdoor attacks were injected into the target model. The target model architecture was ResNet-18, and each dataset was trained. For each attack type, the accuracy of the model was first evaluated on clean and toxic data, then applied to deep learning algorithms to remove backdoors and reevaluate the model's performance. The accuracy of the model on clean and toxic data was used as an evaluation metric.

Table 2 summarizes the performance comparison of the models before and after the backdoor removal. The table shows the accuracy of clean data and poison data.

Table 2. Performance comparison before and after deep unlearning

Attack type	Dataset	Clean data (before)	Standard deviation (before)	Poison data (before)	Standard deviation (before)	Clean data (after)	Standard deviation (after)	Poison data (after)	Standard deviation (after)
BadNet	GTSRB	92.3%	1.2%	98.5%	0.8%	91.7%	1.5%	24.8%	3.0%
Blended attack	GTSRB	90.1%	1.3%	97.8%	0.9%	89.9%	1.6%	22.4%	3.2%
Trojan attack	GTSRB	89.4%	1.4%	99.0%	0.7%	88.8%	1.7%	26.3%	2.8%
BadNet	CIFAR-10	91.2%	1.1%	98.7%	0.6%	90.3%	1.4%	23.1%	2.9%
Blended attack	CIFAR-10	89.8%	1.2%	97.5%	0.8%	89.1%	1.5%	19.2%	3.1%
Trojan attack	CIFAR-10	88.7%	1.3%	98.3%	0.7%	88.2%	1.6%	21.7%	3.0%
BadNet	MNIST	98.3%	0.5%	99.2%	0.4%	97.6%	0.6%	10.5%	2.5%
Blended attack	MNIST	97.9%	0.6%	98.6%	0.5%	97.5%	0.7%	8.9%	2.3%
Trojan attack	MNIST	97.5%	0.7%	99.0%	0.4%	97.2%	0.8%	10.1%	2.4%

It is evident from Table 2 that the deep de-learning method significantly reduced the accuracy of poison data, indicating that the backdoor was successfully removed. After the backdoor was

removed, however, the accuracy of the clean data decreased slightly, indicating that the algorithm affected the normal performance of the model to some extent. Before the backdoor was removed, the model's performance on clean data remained high, with accuracy typically between 89% and 98%. After applying deep unlearning, the accuracy of clean data decreased slightly, by 0.2% to 2.5%. For example, on CIFAR-10 with BadNet, the accuracy of clean data dropped from 91.2% to 90.3%.

After the backdoor was removed, the accuracy of poison data dropped dramatically, especially in complex attack scenarios such as trojan attacks and hybrid attacks. Initially, the model was close to 99% accurate on toxic data, but after applying deep de-learning, the accuracy dropped significantly to 10–26%. On GTSRB, for example, BadNet's accuracy dropped from 98.5% to 24.8% after the backdoor was removed.

The experimental results demonstrated that the proposed method achieved high detection accuracy and adaptability against various types of backdoor attacks (such as BadNet, hybrid attacks, and trojan attacks), and on multiple datasets (MNIST, CIFAR-10, and GTSRB). These results verified the effectiveness and significant contribution of each innovation. Specifically, the design of the lightweight baseline model significantly reduced the computational cost of detection, while still maintaining comparable performance with the target model in resource-constrained environments—providing a reliable reference framework for subsequent sensitivity detection. The sensitivity detection mechanism simulated complex backdoor trigger scenarios through different noise inputs and combined with region-based sensitivity analysis to effectively locate hidden trigger areas, improving detection accuracy by more than 10% when compared with traditional methods. In addition, the deep de-learning algorithm selectively optimized the model parameters in high-risk regions during the backdoor removal process, greatly reducing the model response to the backdoor trigger and ensuring that the classification performance of normal data was only reduced by less than 2%; this significantly reduced the error deletion rate. Overall, the improvements in accuracy, robustness, and computational efficiency validated the critical role of the proposed innovation in model security detection and defense.

Resource Consumption Assessment

To further evaluate the resource efficiency of the proposed method in practical applications, the following were recorded: execution time, memory usage and computational overhead data of the model during training, backdoor detection and backdoor removal. Experiments were conducted on MNIST, CIFAR10 and GTSRB data sets, using two model architectures of ResNet and VGG. Specific data are shown in Table 3.

Table 3. Resource consumption assessment

Data set	Model architecture	Training phase		Backdoor detection phase		Backdoor removal phase	
		Execution time (s)	Memory usage (MB)	Execution time(s)	Memory usage (MB)	Execution time (s)	Memory usage (MB)
MNIST	ResNet18	120	350	15	280	20	300
MNIST	VGG16	150	400	18	300	22	320
CIFAR10	ResNet18	200	450	20	350	25	380
CIFAR10	VGG16	230	500	23	380	28	400
GTSRB	ResNet18	250	550	25	400	30	430
GTSRB	VGG16	280	600	28	430	32	450

From the data in Table 3, it can be observed that under different data sets and model architectures, the proposed method performed well in resource consumption. In the training stage, the execution time and memory occupation were relatively reasonable; the calculation cost was also in an acceptable range. In the backdoor detection and backdoor removal stages, the execution time was shorter, and the increases in memory consumption and computational overhead were minimal. When compared with other methods, the proposed framework demonstrated higher resource efficiency while guaranteeing detection accuracy and the backdoor removal effect—especially in resource-constrained environments.

Real-World Adversarial Scenario Analysis

While controlled experiments validated the framework's effectiveness under synthetic backdoor attacks, real-world adversarial setups introduced additional complexities, such as dynamic trigger evolution, multi-stage attacks, and environmental noise. To bridge this gap, preliminary tests were conducted on a subset of CIFAR-10 images that enhanced real-world damage, such as motion blur and lighting changes. These tests found that semantic sensitivity scores remained valid at detecting triggers embedded in semantically stable regions, such as object contours, when compared with clean conditions. Detection accuracy decreased by $< 5\%$. Highly adaptive attacks that use real-time model output to adjust triggers, however, may require an online semantic update mechanism (Salem et al., 2022), and future work will validate the framework against real-world adversarial datasets and explore adaptive semantic perturbation strategies to deal with dynamic threats.

CONCLUSION AND FUTURE WORK

This study proposed a semantic awareness framework for detecting and eliminating backdoor attacks in deep learning models, addressing the limitations of traditional approaches in dealing with semantic hidden triggers, high computational costs, and interpretability gaps. By integrating semantic web technologies, lightweight baseline model alignment through knowledge distillation, and semantically driven sensitivity analysis, this framework improved detection accuracy by 12% and error deletion rates by 20% during backdoor deletions, while keeping clean data performance degradation below 2%. Experimental validation on MNIST, CIFAR-10, and GTSRB datasets showed that the model was effective in identifying subtle semantic-level triggers and operating efficiently in resource-constrained environments. These findings highlighted the critical role of semantic analytics in enhancing AI security, with direct implications for safety-critical applications such as medical diagnostics and autonomous driving, where reliable detection of hidden backdoors is critical to preventing catastrophic failures.

Based on this research, future work will focus on four actionable directions to extend the applicability and robustness of the framework. First, the framework will be adapted to emerging AI paradigms, such as federated learning and large language models. In federated learning, decentralized training across multiple clients presents challenges in detecting embedded backdoors in collaborative model updates. Future studies may include developing cross-client semantic conformance checks, where semantic ontologies are used to monitor differences in feature contributions between participants, thus enabling early detection of malicious updates that exploit semantic ambiguity. For large language models, linguistic knowledge graph integrations can be explored to simulate context in prompts and detect backdoor triggers masquerading as legitimate language patterns, such as subtle phrase manipulation. These explorations will include designing semantic-aware hint filters to identify abnormal activation patterns in the transformer layer, ensuring robustness to stealth backdoor hints targeting conversational systems or code-generation models.

Second, adaptive semantic perturbation strategies will be developed to deal with evolving attack dynamics. Current approaches rely on static noise patterns, which may not be resilient to adaptive attacks based on model responses to dynamically tuned triggers (Salem et al., 2022). Meta-learning or adversarial training will be utilized to generate context-aware perturbations that adapt to real-world

noise—domain-specific textures in medical imaging or environmental changes in autonomous driving, for example. By incorporating semantic metadata into perturbation generation, such as object-class hierarchies and spatial relationships, the framework can dynamically amplify subtle trigger semantic mismatches, improving resilience to adaptive attacks while also maintaining interpretability through explicit semantic mapping.

Third, this framework will be extended to multimodal and cross-modal environments in which backdoor triggers may exploit the gap between visual, text, and acoustic modes—for example, audiovisual triggers in smart home devices. Cross-modal semantic alignment metrics, such as joint visual–text semantic embedding, will be designed to detect inconsistencies in multimodal feature interactions. In an autonomous vehicle, for example, the differences between visual traffic sign features and the corresponding text labels in the semantic knowledge graph can be monitored, identifying triggers that manipulate cross-modal correspondents. In addition, to enable holistic detection of triggers across multiple data types, future research will explore semantic-driven fusion strategies that integrate heterogeneous mode-specific ontologies.

Fourth, the synergy between semantic awareness and lightweight model optimization will be examined for edge and internet of things devices. While the current framework reduces the computational overhead with a lightweight baseline model, further optimization for resource-constrained settings is required. Semantically guided model pruning will be explored, wherein redundant parameters unrelated to higher semantic features are selectively removed—this strikes a balance between detection accuracy and inference efficiency. In addition, future studies will integrate hardware-software co-design with trusted execution environments, such as ARM TrustZone, to enforce semantic-level isolation of model components, thereby ensuring secure execution while minimizing latency for real-time applications.

These directions aim to strengthen AI security in diverse and increasingly complex machine learning ecosystems, ensuring confidence in the face of complex semantic-level threats. By connecting semantic reasoning with adaptive machine learning and interdisciplinary techniques, this framework can be developed into a general-purpose solution for protecting the next generation of AI systems from complex context-aware attacks.

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